Real-World Applications & Global Change Projections

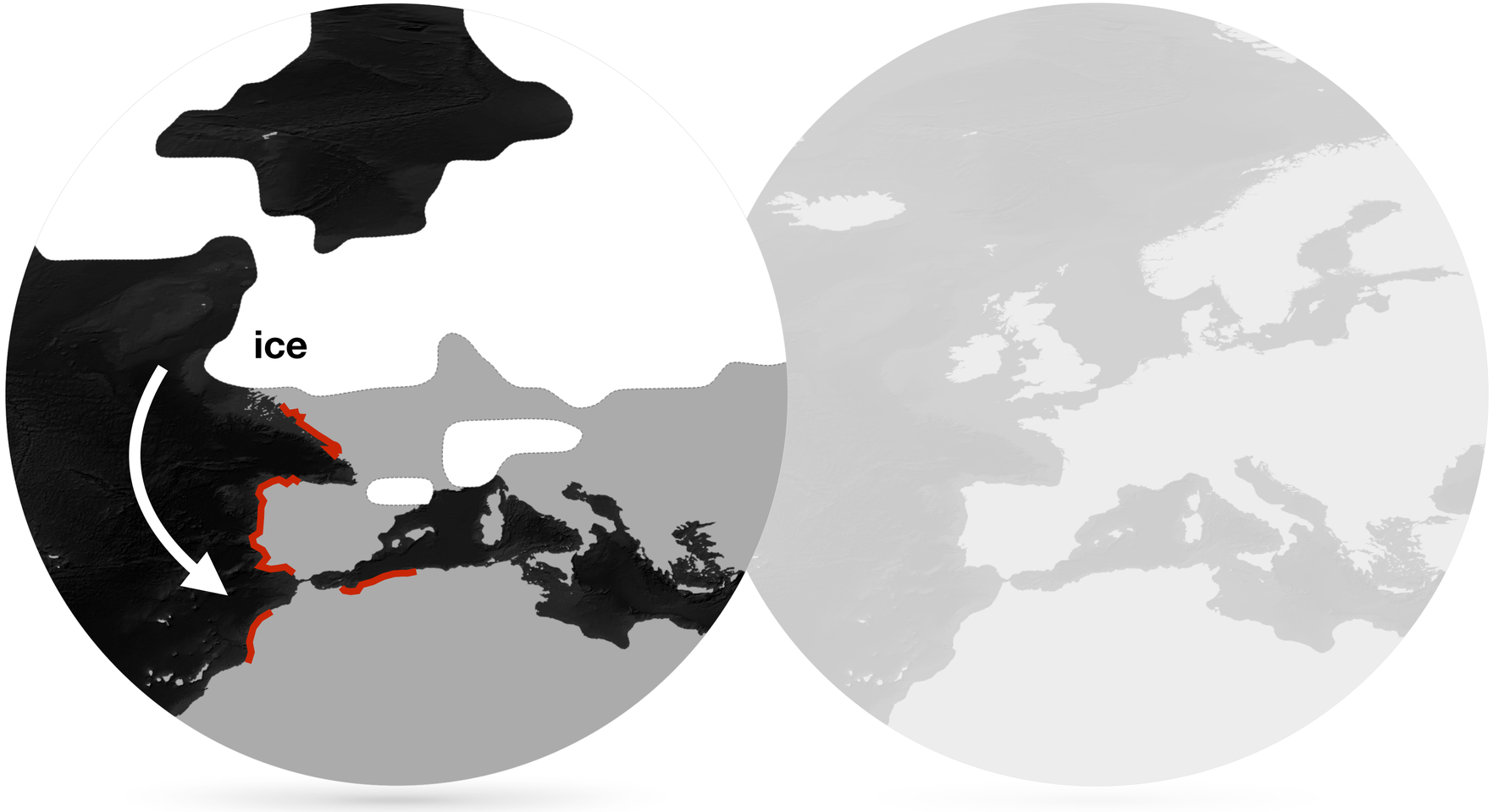
Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

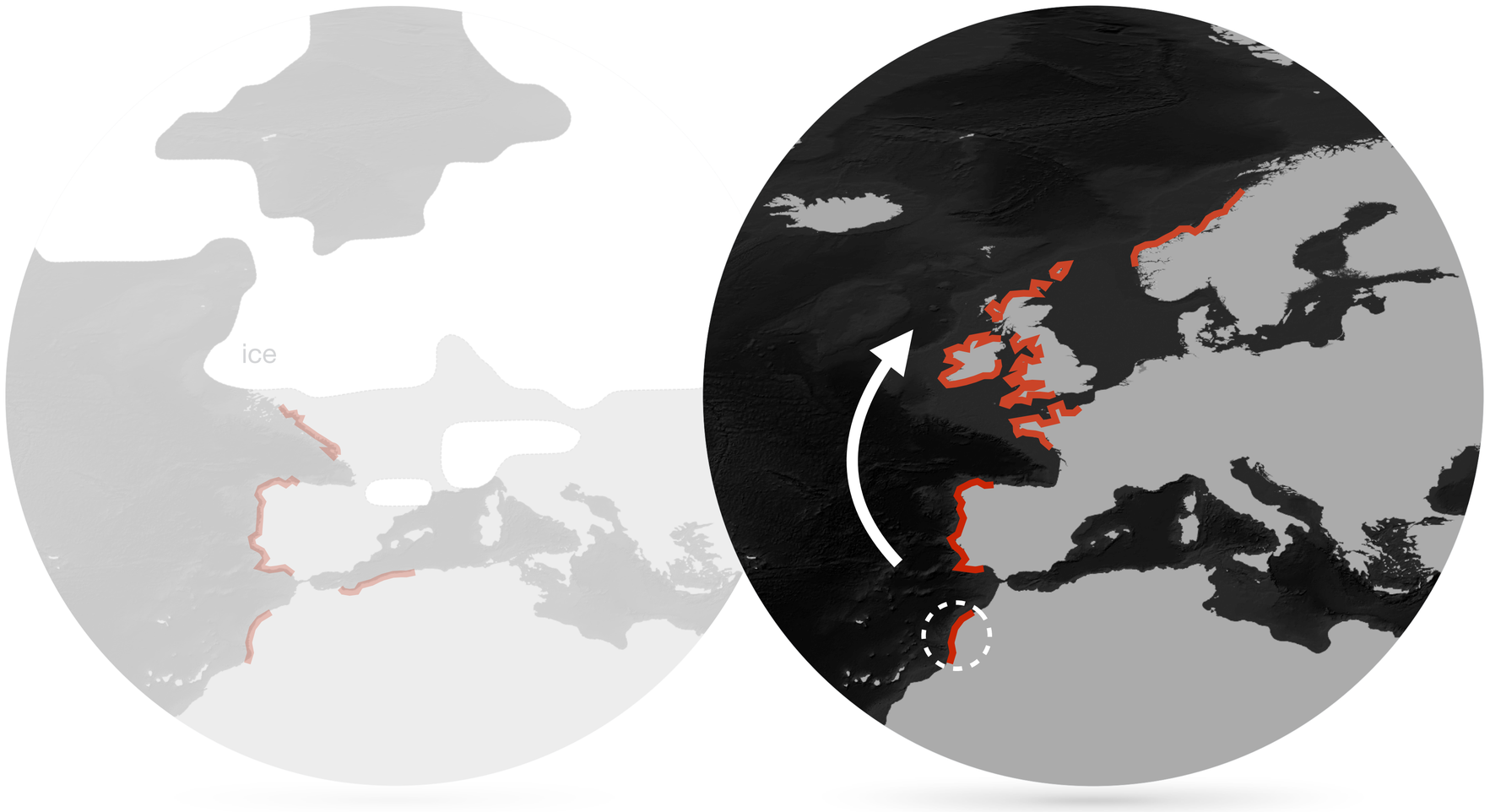
Hindcasting to Past Climates to Explain Genetic Diversity

The paradigm of past range shifts structuring the distribution of intra-specific biodiversity is well established, but it lacks replication.

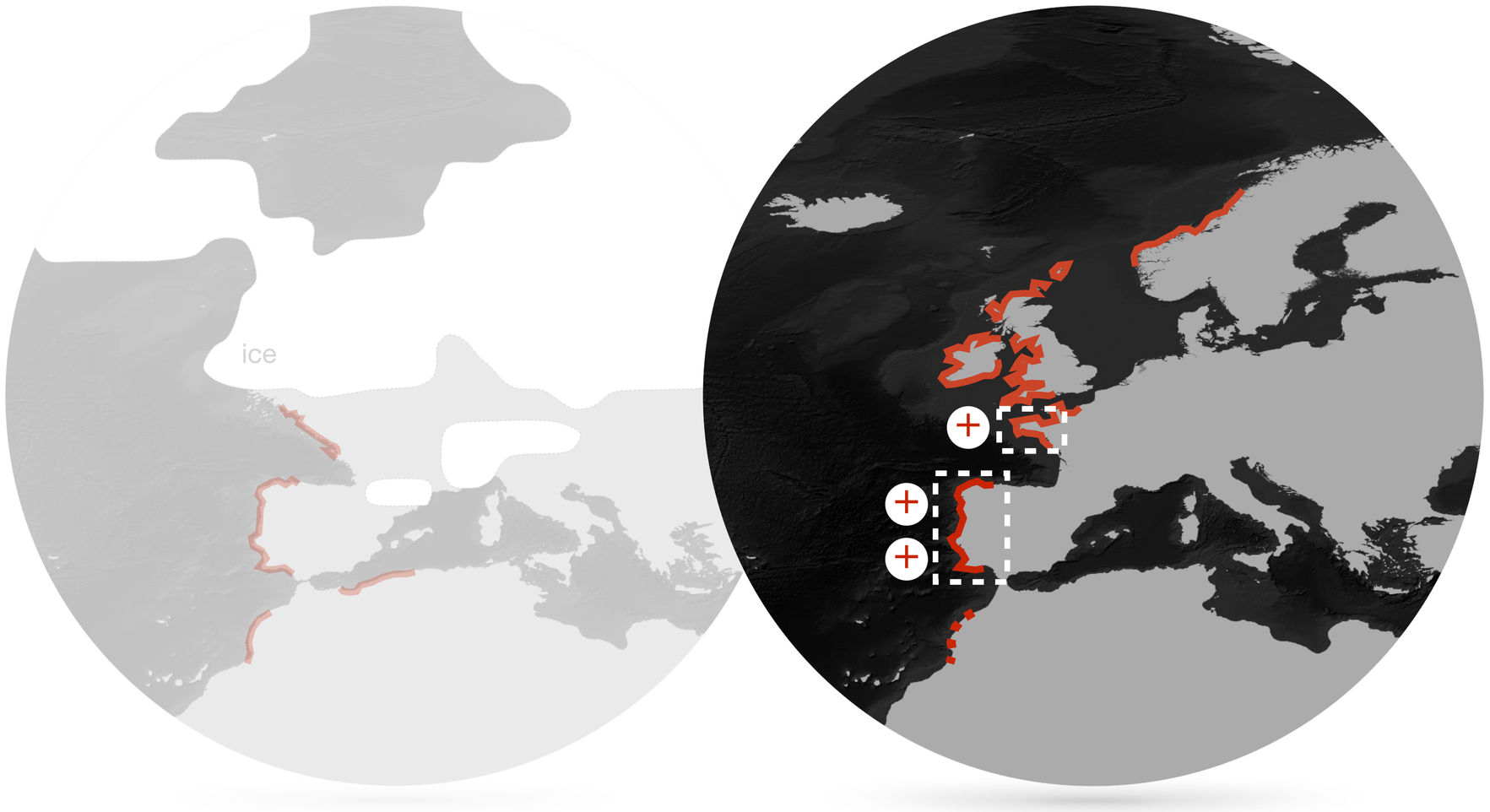
Studies addressing the topic mostly in the NE Atlantic, and **missing biological models exposed to** comparable conditions in **independent geographic regions**.



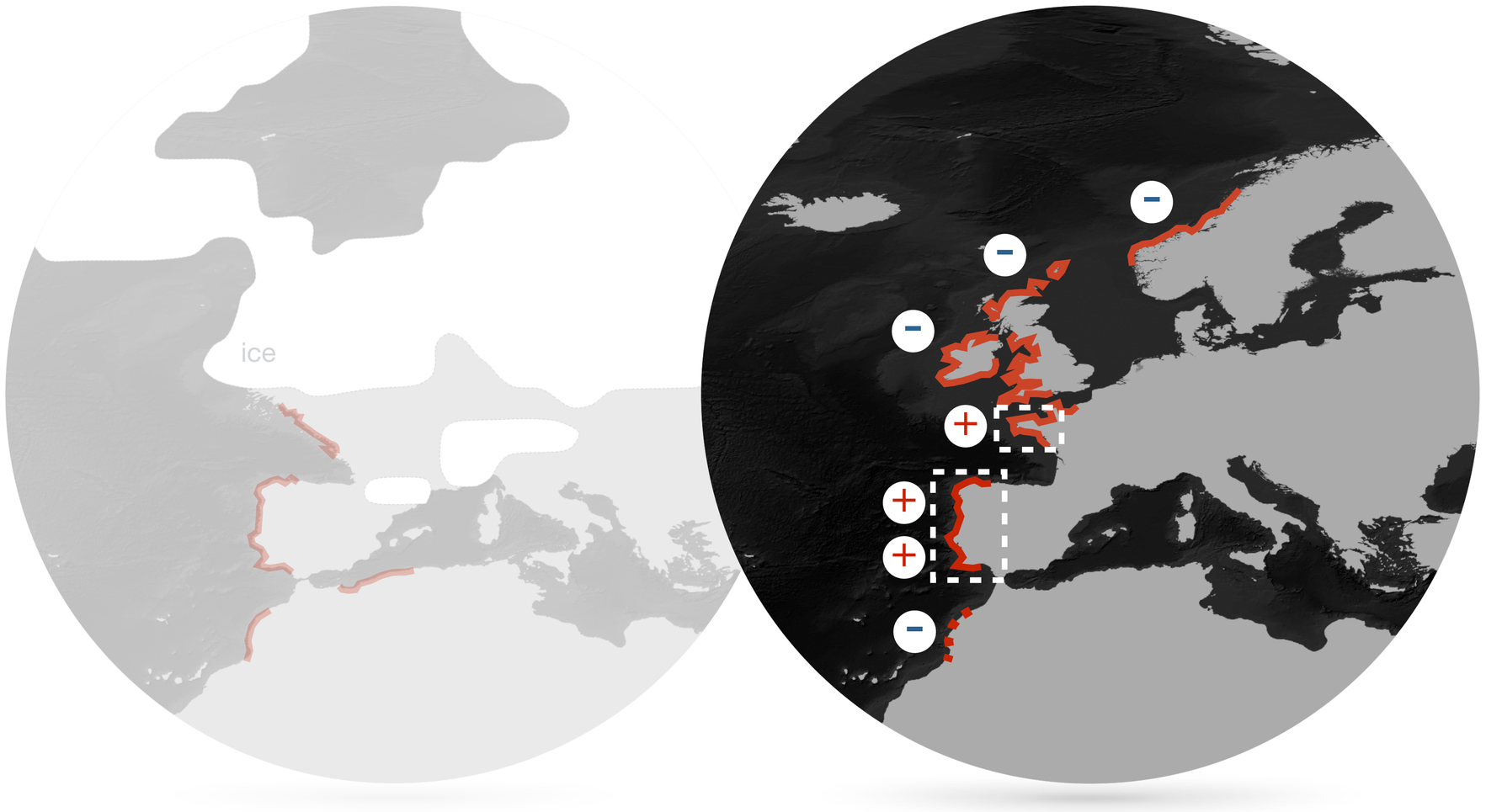
During glacial periods (e.g., LGM), the advance of massive ice sheets and the colder climate led **species to contract their high latitude distributions**.



During interglacials, the warmer conditions **allowed species to (re)colonise higher latitudes**. **Extinctions might have occurred at low latitude range margins**.



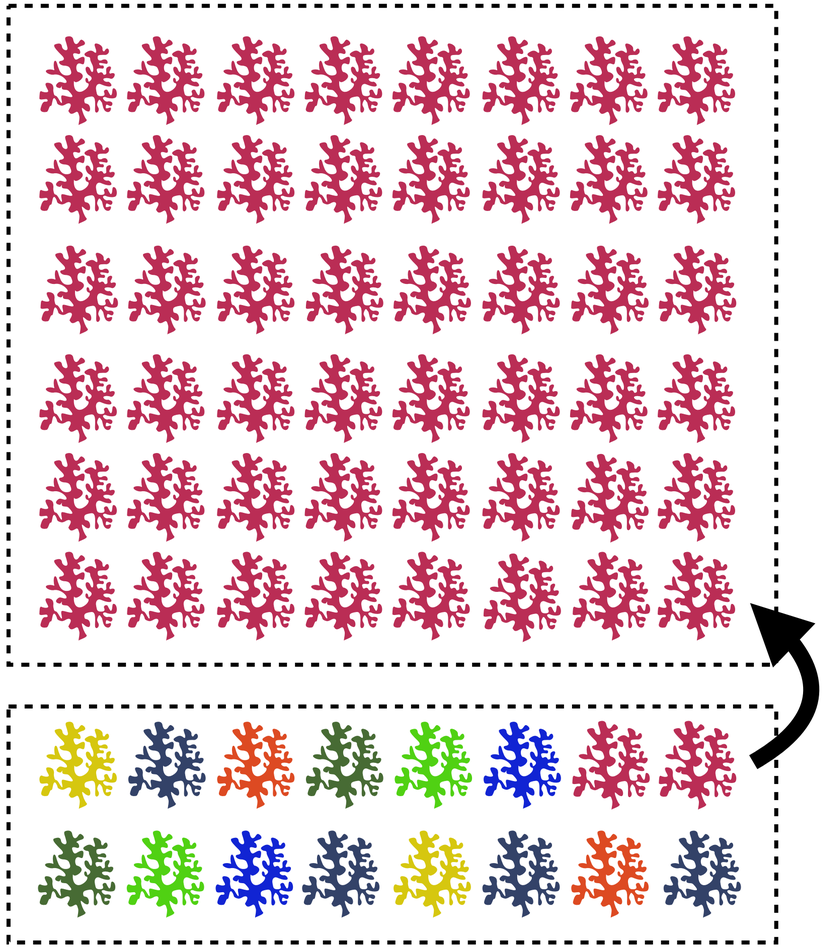
Range shifts structured regions of **long-term persistence (i.e.,climatic refugia) where populations had time to accumulate higher and unique genetic diversity**.



Where local extinctions or expansions occurred, **lower genetic diversity is expected**, due to bottlenecks on effective population sizes or founder effects.

Expansion front (e.g., Post-LGM)

Low and homogeneous diversity



Long-term refugia

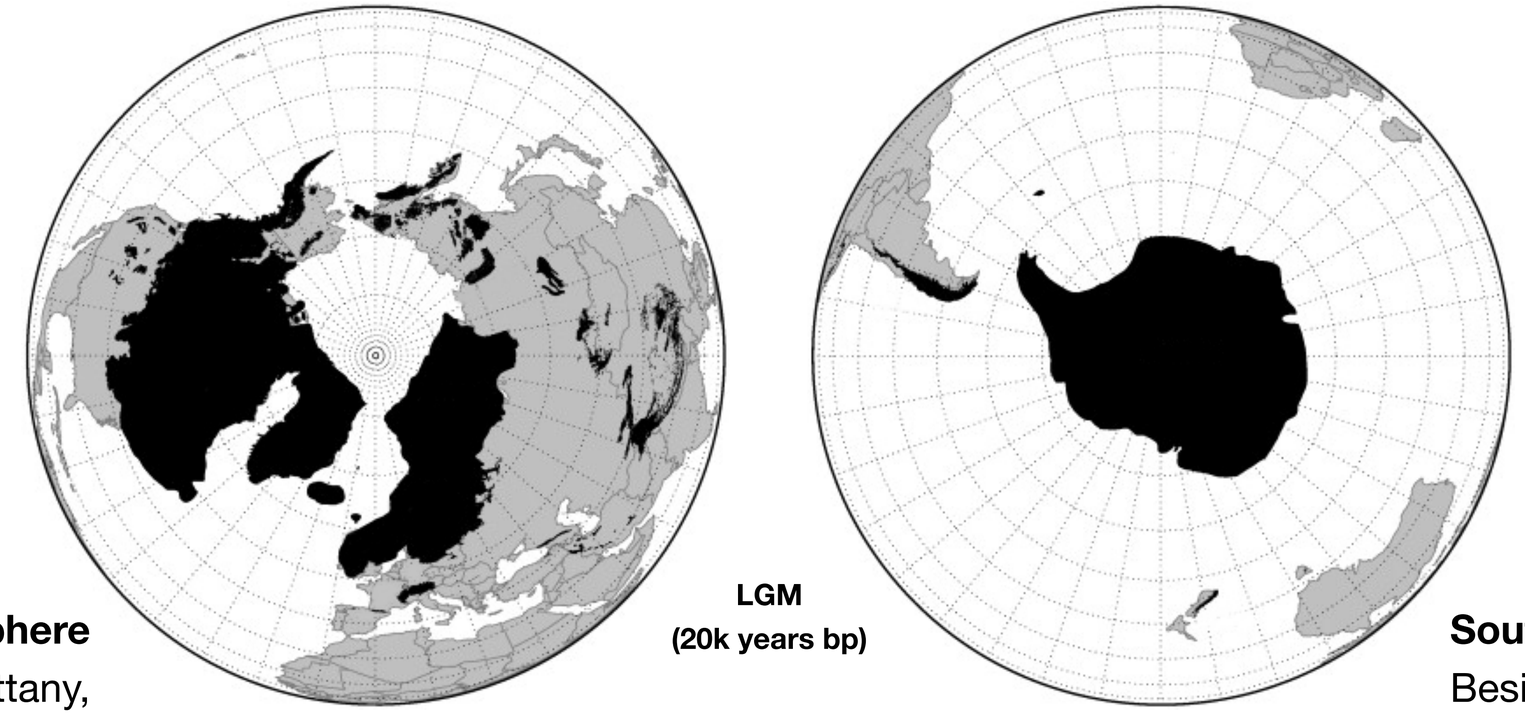
High and unique diversity

Founder effects **tend to leave reduced diversity** because colonizations tend to be performed by the **spread of few individuals / alleles**.

The paradigm of past range shifts structuring the distribution of intra-specific biodiversity is well established, but it lacks replication.

Studies addressing the topic mostly in the NE Atlantic, and **missing biological models exposed to** comparable conditions in **independent geographic regions**.

Northern hemisphere
Ice advances down to Brittany, Maine and Vancouver Island



LGM
(20k years bp)

Southern hemisphere
Besides Antarctica, only Patagonia was affected

During glacial periods (e.g., LGM), the **higher latitudes of the two hemispheres provided very distinct conditions** for evolution.

** such contrasting conditions might have caused differences in the genetic diversity drivers between hemispheres*



Global distribution



The **giant kelp *Macrocystis pyrifera* is an ideal model** to explore the topic considering its **broad antitropical distribution** with **two independent replicate conditions exposed to extreme glacial conditions**.

Genetic diversity estimated across the entire species distribution.

(115 sites from Alaska to the Southern Ocean)

Genetic diversity levels **compared to SDM predicting range shifts** (extinctions and expansions) **and climatic refugia** (persistence) **from the LGM to the present**.

Assis et al., 2023

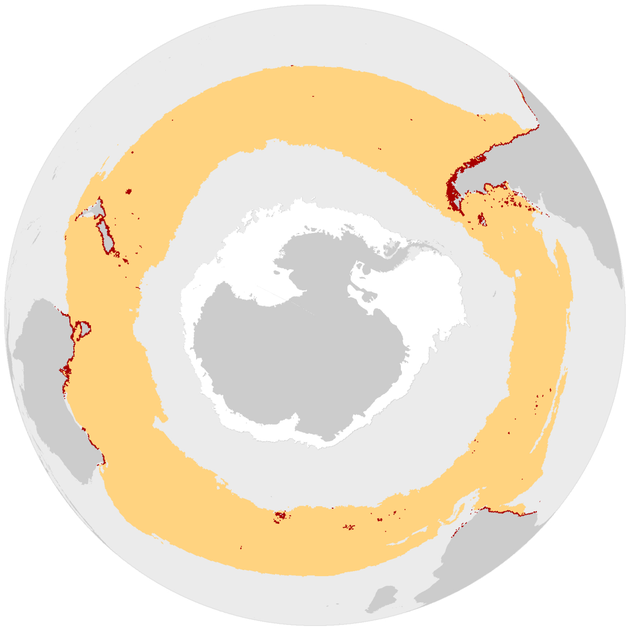
biodiversityDS.

Pacific Ocean

Southern Ocean

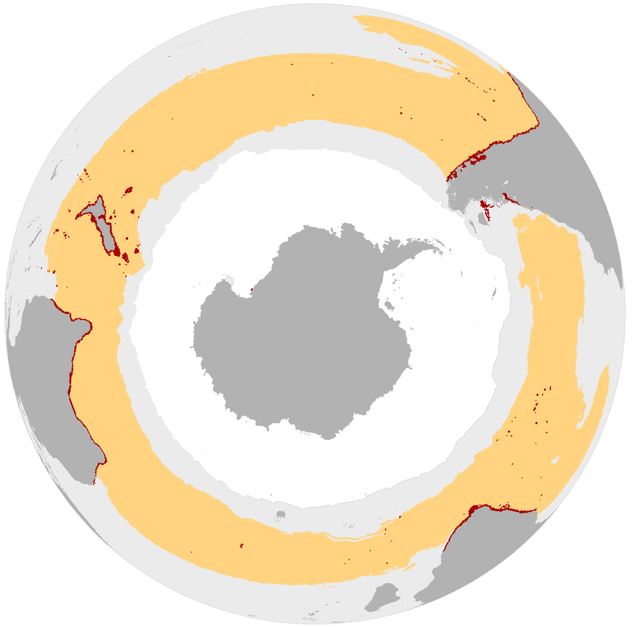
Present-day distribution





Past distribution [LGM]





SDM predictions

suitable habitats

Pacific Ocean

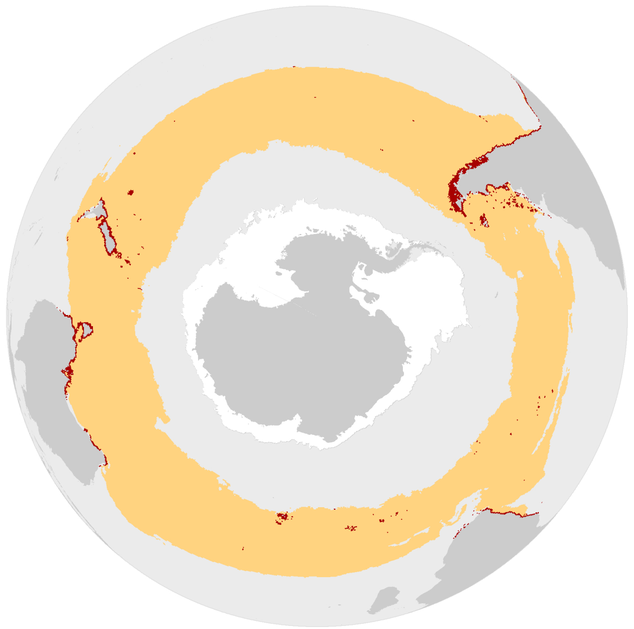
Southern Ocean

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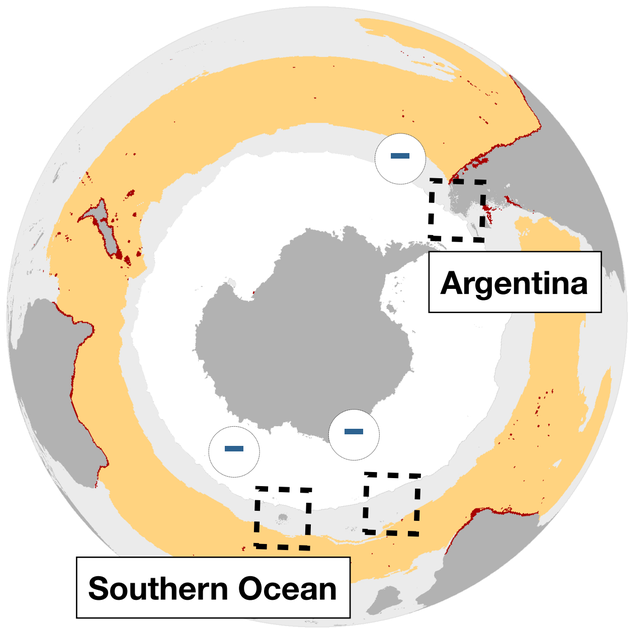
Present-day distribution

Past distribution [LGM]









During LGM period

Broad **extinctions at higher latitudes**, leading to **40% range contraction**, globally.

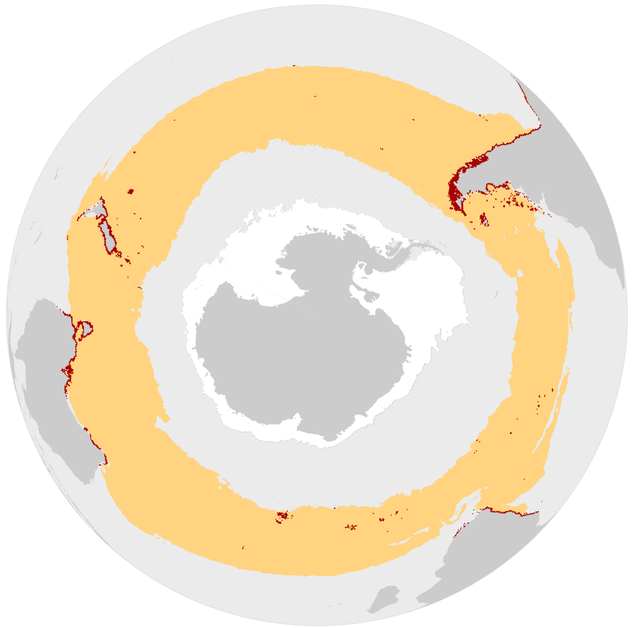
Pacific Ocean

Southern Ocean

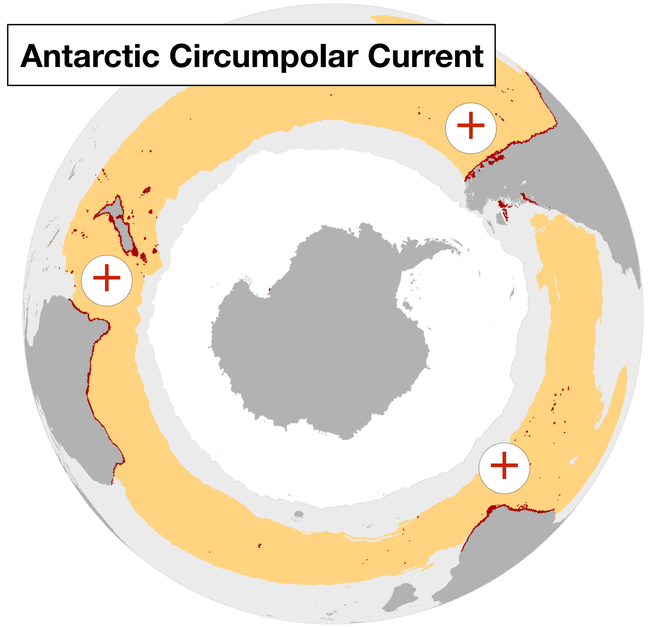
Present-day distribution

Past distribution [LGM]





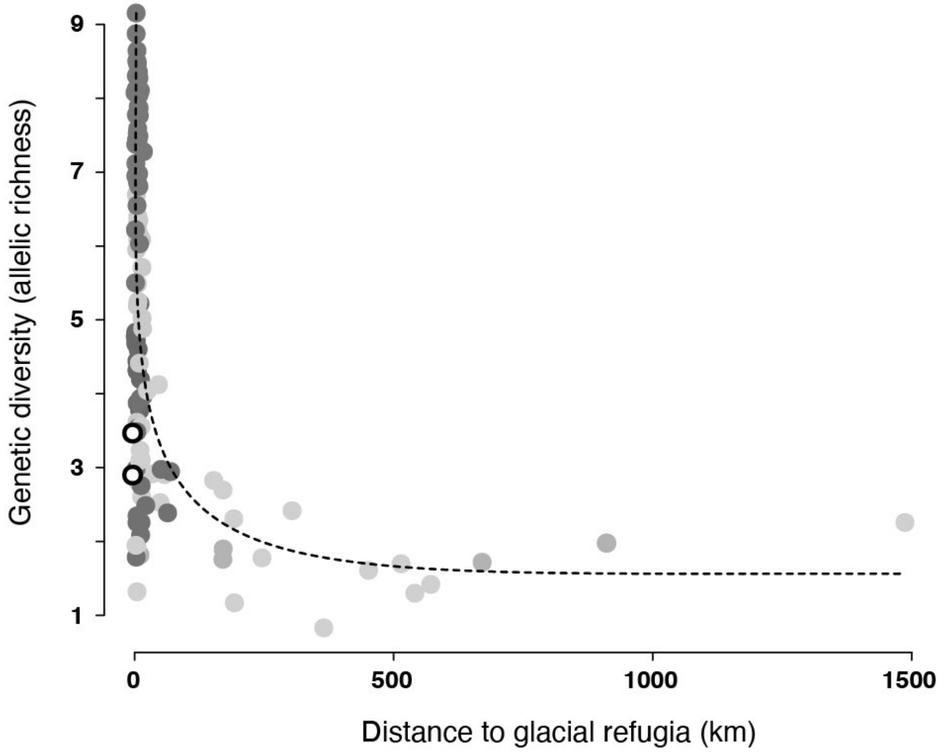


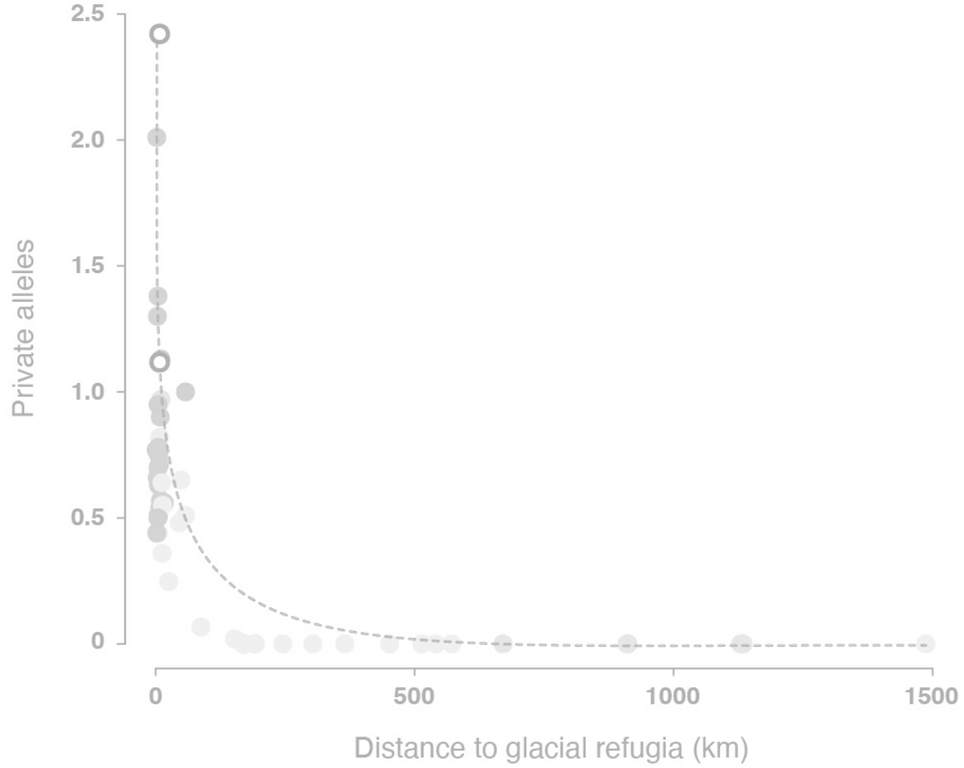


Comparing time periods

Multiple climatic refugia, where the species could have persisted.

Comparison of genetic diversity vs. distance to climatic refugia (both hemispheres)

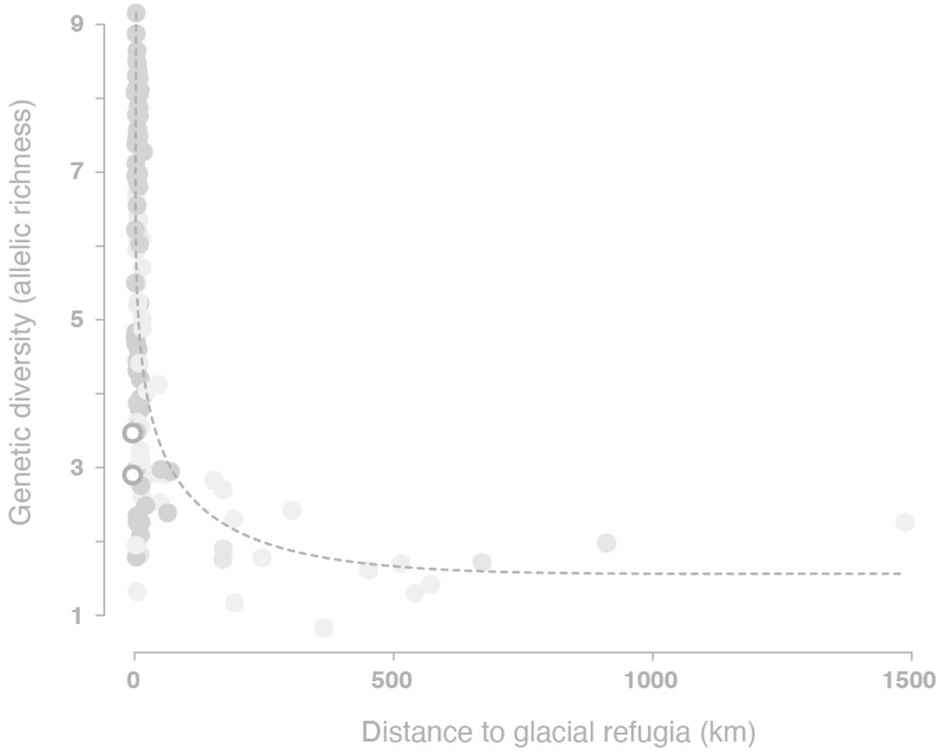


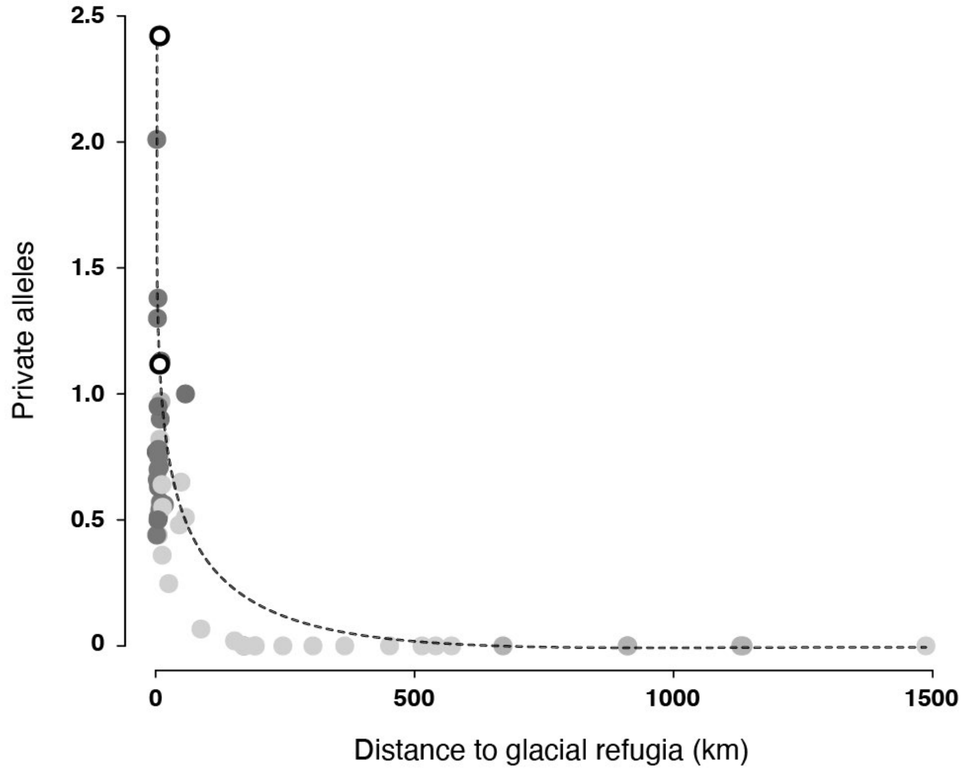


Diversity systematically lower outside refugia (regions of persistence).

Diversity up to 3-fold lower at distances higher than 75 km of refugia.

Comparison of genetic diversity vs. distance to climatic refugia (both hemispheres)





Diversity systematically lower outside refugia (regions of persistence).

Endemic alleles (unique of populations) were **<1 at distances higher than 50 km of refugia**, and zero at 150 km of refugia (50 populations with no endemic diversity).

By coupling SDM hindcasts with empirical genetic data: Genetic diversity levels can be shaped by past climate oscillations

(first model with replication across the hemispheres).

Higher latitudes

Post-glacial expansions left populations with reduced and homogenous diversity.

Lower latitudes

Range shifts structured **long-term refugia, with populations harboring higher and unique diversity**.